Representing Linear Maps

Matrix Transformation	Linear Transformation
$\vec{v} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in Mat_{n\times 1}(F)$	$v = (a_1, \dots, a_n) \in F^n$
$M \in Mat_{m\times n}(F)$	$L: F^n \to F^m$ s. t. $L(\lambda_1 v + \lambda_2 w) = \lambda_1 L(v) + \lambda_2 L(w)$
$\vec{v} \mapsto M\vec{v} \in Mat_{m\times 1}(F)$	
E.g. $\vec{v} \in \mathbb{R}^2 \mapsto R_\theta \vec{v} \in \mathbb{R}^2$	E.g. $L: \mathbb{Z}_3^2 \to \mathbb{Z}_3^3$ $L(x,y) = (x+y, 2y, x+2y)$.

Matrix Transformation $\rightsquigarrow$ Linear Transformation

Given $M \in Mat_{m\times n}(F)$, we can define a map $L_M : F^n \to F^m$ as follows:

$$L_M(x_1, x_2, \dots, x_n) = (y_1, \dots, y_m) \Leftrightarrow M\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

<u>Example.</u> If $M = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix} \in Mat_{2\times 3}(\mathbb{Z}_3)$, then

$$\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x+y \\ x+2y+z \end{pmatrix}$$

is the matrix transformation, the associated map is $L_M : \mathbb{Z}_3^3 \to \mathbb{Z}_3^2$,

$$L(x,y,z) = (2x+y,\ x+2y+z).$$

<u>Proposition.</u> For $M \in \mathrm{Mat}_{m\times n}(F)$, the associated map $L_M : F^n \to F^m$ is linear.

Proof. By definition, if $\vec{v} = (v_1, \dots, v_n)$, $\vec{w} = (w_1, \dots, w_n)$, then

$$\begin{aligned}
L_M(\lambda\vec{v} + \lambda'\vec{w}) &= L_M(\lambda v_1 + \lambda' w_1, \dots, \lambda v_n + \lambda' w_n) \\
&= M\begin{pmatrix} \lambda v_1 + \lambda' w_1 \\ \vdots \\ \lambda v_n + \lambda' w_n \end{pmatrix} \\
&= M\left(\lambda\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \lambda'\begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}\right) \\
&= \lambda M\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \lambda' M\begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \\
&= \lambda L_M(\vec{v}) + \lambda' L_M(\vec{w})
\end{aligned}$$

Linear Transformation $\rightsquigarrow$ Matrix Transformation

Proposition. If $L: F^n \to F^m$ is linear, then there exists a unique $M \in Mat_{m\times n}(F)$ representing L.

Proof. Set $\vec{e}_i$ the ith standard basis vector. Define notation $L(\vec{e}_i) = (c_{1i}, c_{2i}, \dots, c_{mi}) \in F^m$. Consider the matrix

$$M_L = \begin{pmatrix} c_{11} & \cdots & c_{1n} \\ \vdots & \ddots & \vdots \\ c_{m1} & \cdots & c_{mn} \end{pmatrix} \in Mat_{m\times n}(F)$$

ith column $\leftrightarrow L(\vec{e}_i)$

Claim: M_L represents L.

Let $\vec{v} = (v_1, \dots, v_n) \in F^n$. We can write

$$\vec{v} = v_1\vec{e}_1 + v_2\vec{e}_2 + \cdots + v_n\vec{e}_n.$$

Since L is linear,

$$\begin{aligned} L(\vec{v}) &= L(v_1\vec{e}_1 + \cdots + v_n\vec{e}_n) \\ &= v_1 L(\vec{e}_1) + v_2 L(\vec{e}_2) + \cdots + v_n L(\vec{e}_n) \\ &= v_1(c_{11}, \dots, c_{m1}) + \cdots + v_n(c_{1n}, \dots, c_{mn}) \end{aligned}$$

$$= (c_{11}v_1 + \cdots + c_{1n}v_n, \ldots, c_{m1}v_1 + \cdots + c_{mn}v_n)$$

Now notice:

$$M\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix}\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

$$= \begin{pmatrix} c_{11}v_1 + \cdots + c_{1n}v_n \\ \vdots \\ c_{m1}v_1 + \cdots + c_{mn}v_n \end{pmatrix}$$

$\therefore$ M represents L as desired.

<u>Intuitively</u>. jth column of $M \Leftrightarrow L(\vec{e}_j)$.

<u>Example</u>. If $L: \mathbb{R}^4 \to \mathbb{R}^3$ is given by

$$L(x, y, z, w) = (5x + 2y - z + 3w,\ x - y + 2z,\ x + y - 4w)$$

then $L(\vec{e}_1) = L(1,0,0,0) = (5, 1, 1)$,

$L(\vec{e}_2) = L(0,1,0,0) = (2, -1, 1)$

$L(\vec{e}_3) = L(0,0,1,0) = (-1, 2, 0)$

$L(\vec{e}_4) = L(0,0,0,1) = (3, 0, -4)$

Thus, L is represented by $M = \begin{pmatrix} 5 & 2 & -1 & 3 \\ 1 & -1 & 2 & 0 \\ 1 & 1 & 0 & -4 \end{pmatrix}$

Representing Compositions

Theorem. If $L_1 : F^n \to F^m$ is represented by $M_1 \in \mathrm{Mat}_{m\times n}(F)$ and $L_2 : F^m \to F^p$ is represented by $M_2 \in \mathrm{Mat}_{p\times m}(F)$, then $L_2 \circ L_1 : F^n \to F^p$ is represented by $M_2 M_1$.

Example. $L_1 : \mathbb{Z}_5^2 \to \mathbb{Z}_5^3$, $L_2 : \mathbb{Z}_5^3 \to \mathbb{Z}_5^3$ with

$$L_1(x,y) = (x+2y,\ 3x+4y,\ y),$$
$$L_2(x,y,z) = (y+4z,\ x+y+3z,\ y+z)$$

hand composition

$$(L_2 \circ L_1)(x,y) = (3x+3y,\ 4x+4y,\ 3x)$$

Now,

$$L_1 \rightsquigarrow M_1 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{pmatrix},$$

$$L_2 \rightsquigarrow M_2 = \begin{pmatrix} 0 & 1 & 4 \\ 1 & 1 & 3 \\ 0 & 1 & 1 \end{pmatrix}.$$

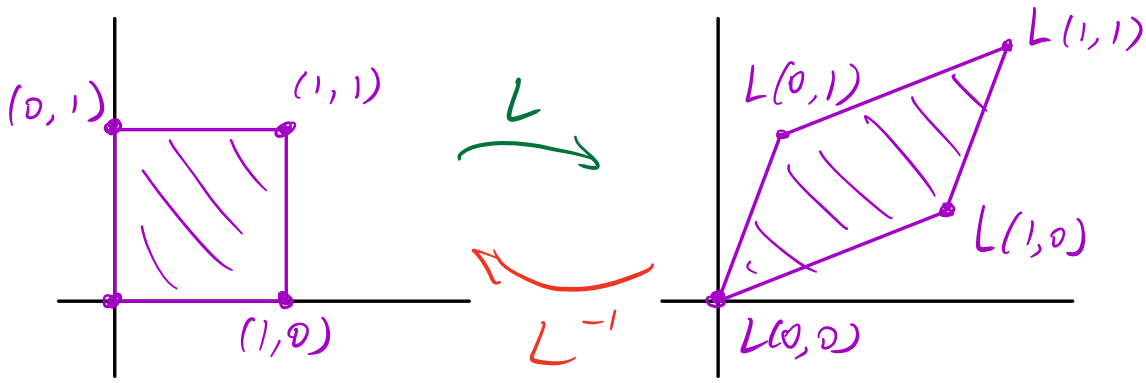
$$M_2 M_1 = \begin{pmatrix} 3 & 3 \\ 4 & 4 \\ 3 & 0 \end{pmatrix}.$$

Theorem. If $L: F^n \to F^n$ is invertible, and L is represented by M, then

① M is invertible
② L^{-1} is represented by M^{-1}.

Invertible Maps on $\mathbb{R}^2$

We only consider $L: \mathbb{R}^2 \to \mathbb{R}^2$ invertible. As before, $L(x,y) = xL(1,0) + yL(0,1)$, and $L(0,0) = (0,0)$, $L(1,1) = L(1,0) + L(0,1)$.



Theorem. If $L: \mathbb{R}^2 \to \mathbb{R}^2$ is represented by M, then the signed area of the image of the unit square under L is $\det M$.

Eigenvalues and Eigenvectors

<u>Definition.</u> Let $M \in Mat_n(F)$. Let $\vec{v}$ be a nonzero column vector in F^n. We say that $\vec{v}$ is an eigenvector for M if there is some scalar λ such that

$$M\vec{v} = \lambda\vec{v}.$$

If so, we say λ is an eigenvalue of M and $\vec{v}$ is an eigenvector associated to λ.

In general, for $M \in Mat_n(F)$, $\lambda \in F$, we have

$$M\vec{v} = \lambda\vec{v} \Rightarrow M\vec{v} - \lambda\vec{v} = \vec{0}$$
$$\Rightarrow (M - \lambda I)\vec{v} = \vec{0}$$

<u>Definition (Eigenspace).</u>

The λ-eigenspace of M is the set of solutions

$$\{\vec{v} \mid (M - \lambda I) = \vec{0}\}.$$

<u>Example.</u> Find the $\lambda=2$-eigenspace over $\mathbb{Z}_7$ for

$$M = \begin{pmatrix} 4 & 6 & 6 \\ 2 & 1 & 6 \\ 2 & 6 & 1 \end{pmatrix}.$$

<u>Solution</u>

$$M - \lambda I = M - 2I = \begin{pmatrix} 2 & 6 & 6 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}$$

$$\sim \begin{pmatrix} 2 & 6 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & 3 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

This corresponds to

$$x_1 + 3x_2 + 3x_3 = 0$$
$$\Rightarrow \quad x_1 = 4x_2 + 4x_3.$$

The solution set is

$$\left\{ \begin{pmatrix} 4s+4t \\ s \\ t \end{pmatrix} \middle| \, s,t \in \mathbb{Z}_7 \right\} = \left\{ s\begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} \middle| \, s,t \in \mathbb{Z}_7 \right\}$$

Note: If $\vec{v}, \vec{w}$ are eigenvectors of M with eigenvalue λ, then so are $c\vec{v}$ and $\vec{v}+\vec{w}$ for $c \in F \setminus \{0\}$.

The Characteristic Equation

Proposition. $\lambda \in F$ is an eigenvalue of M iff

$$\det(M - \lambda I) = 0.$$

Proof. ($\Rightarrow$) Suppose λ is an eigenvalue of M. Then there is a nonzero vector $\vec{v}$ such that
$M\vec{v} = \lambda\vec{v} \Rightarrow (M - \lambda I)\vec{v} = \vec{0}$.

Assume $\det(M - \lambda I) \neq 0$. Then $(M - \lambda I)$ is invertible and

$$\vec{v} = (M - \lambda I)^{-1}\vec{0} = \vec{0}.$$

This contradicts, as $\vec{v}$ is nonzero. Thus, $\det(M - \lambda I) = 0$.

($\Leftarrow$). Suppose $\det(M - \lambda I) = 0$. Let $A \sim (M - \lambda I)$ be the RREF for $M - \lambda I$. Since $\det(M - \lambda I) = 0$, $M - \lambda I$ is not invertible, so $A \neq I_n$. Since A is square, RREF, not I_n, it must have a row of zeros. This forces at least one column of A to not have a pivot position (leading

variable).

$\Rightarrow$ There is at least one free variable for the system of equations

$$(M - \lambda I \mid \vec{0}).$$

$\Rightarrow$ $(M - \lambda I)\vec{v} = \vec{0}$ has a nonzero solution.

$\Rightarrow$ There is an eigenvector with $M\vec{v} = \lambda\vec{v}$.